EE3355: SIGNAL AND SYSTEM

LABORATORY 01

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REG No. : EE/2021/4392

SEMESTER: 03

DATE : 20/11 /2026

Table 1.1: Summative Laboratory Form

Semester	03
Module Code	EE3355
Module Name	Signal and system
Lab Number	01
Lab Name	Continuous-time Signal Analysis
Lab conduction date	2026.11.04
Report Submission date	2026.11.20

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1 Observation

1.1 Part 01 Observe the waveform in Time domain and Frequency domain

- Amplitude-5.12V
- Frequency-2.000480kHzkHz
- Peak-to-peak voltage-10.46V
- Period-499.880µs

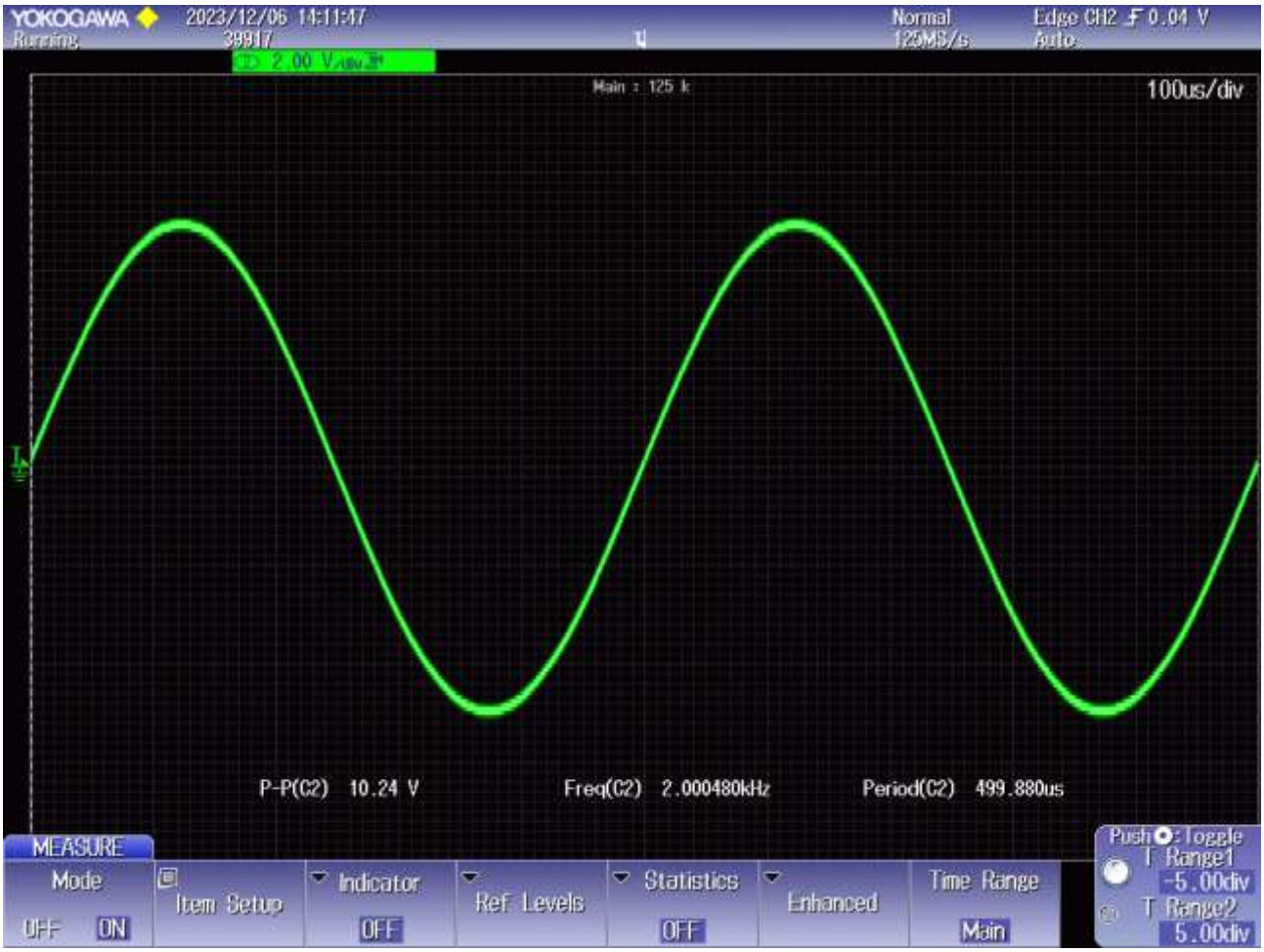


Figure 1 : Observed Time domain waveform using Oscilloscope

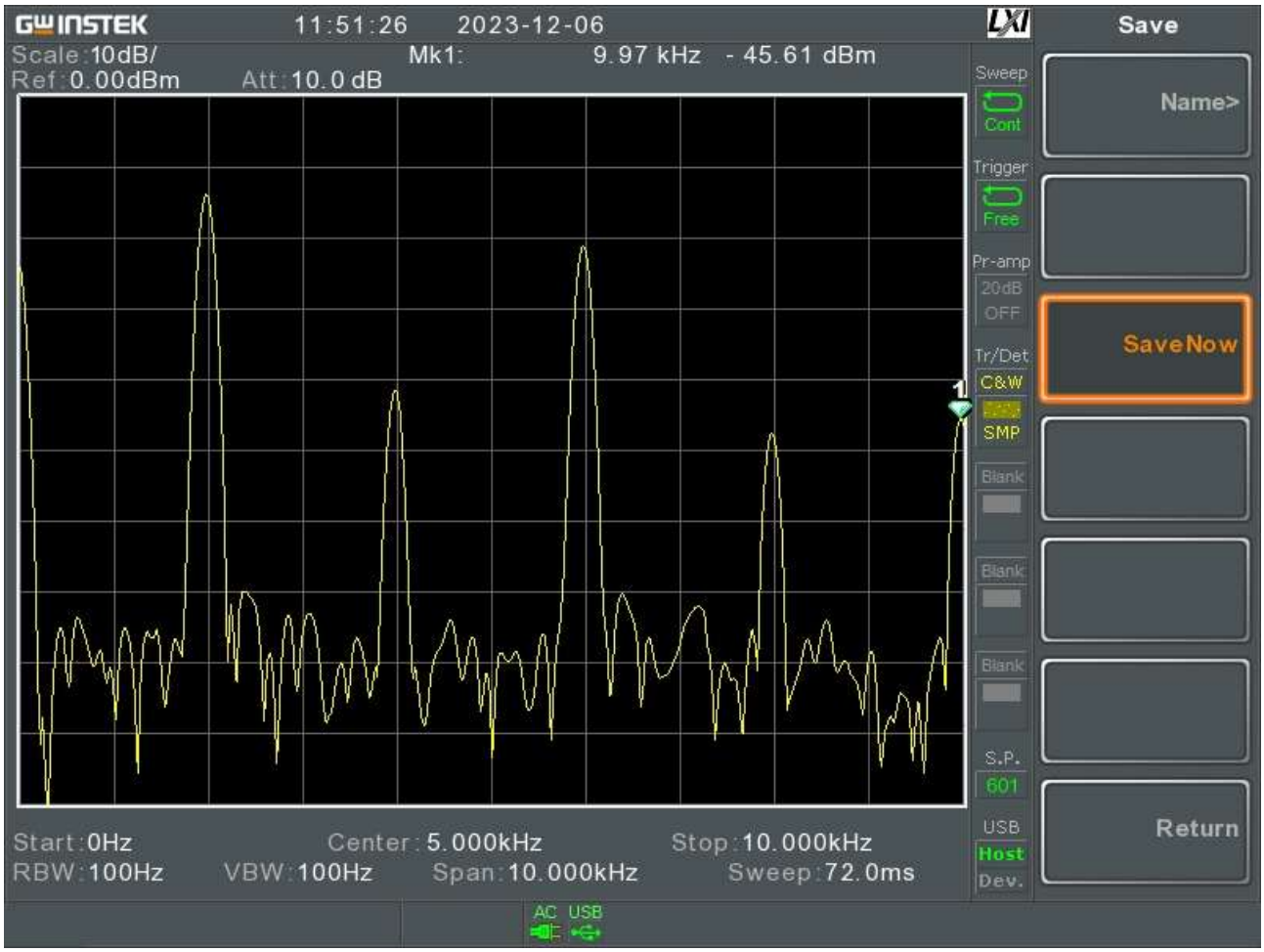


Figure 2 : Observed Frequency domain waveform

1.2 Square waveform

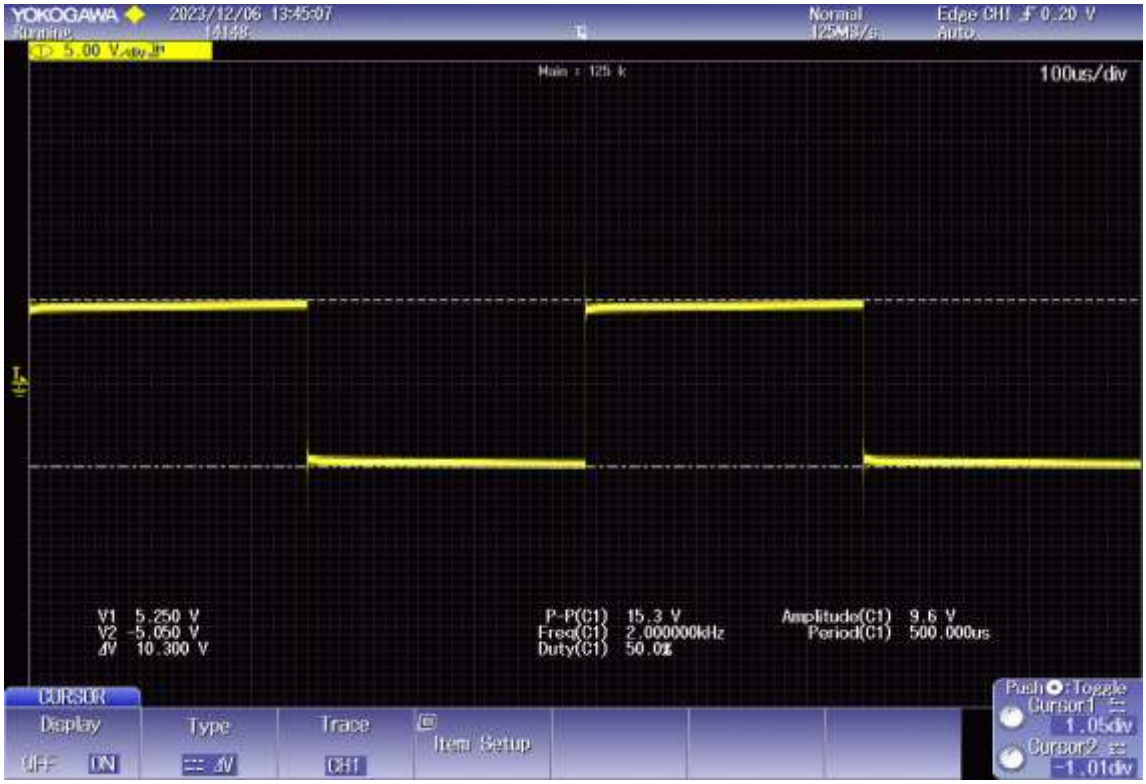


Figure 3 :Observed Time domain square waveform-50% Duty cycle

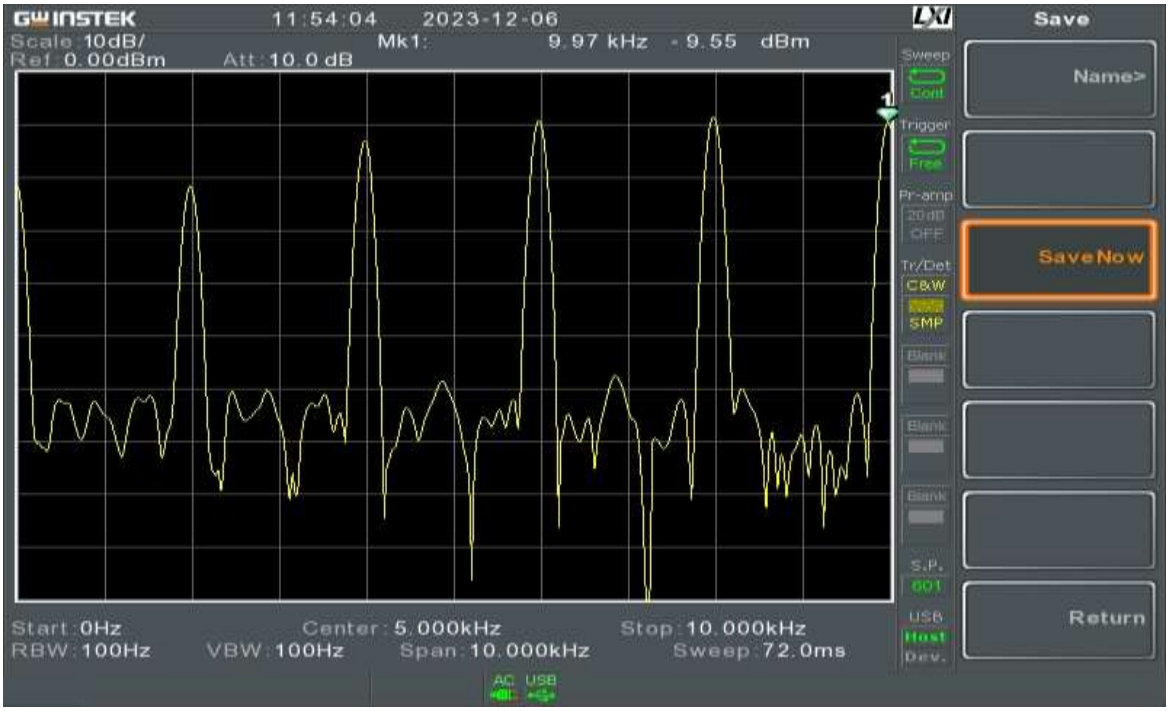


Figure 4 : Observed Frequency domain waveform-50% Duty cycle

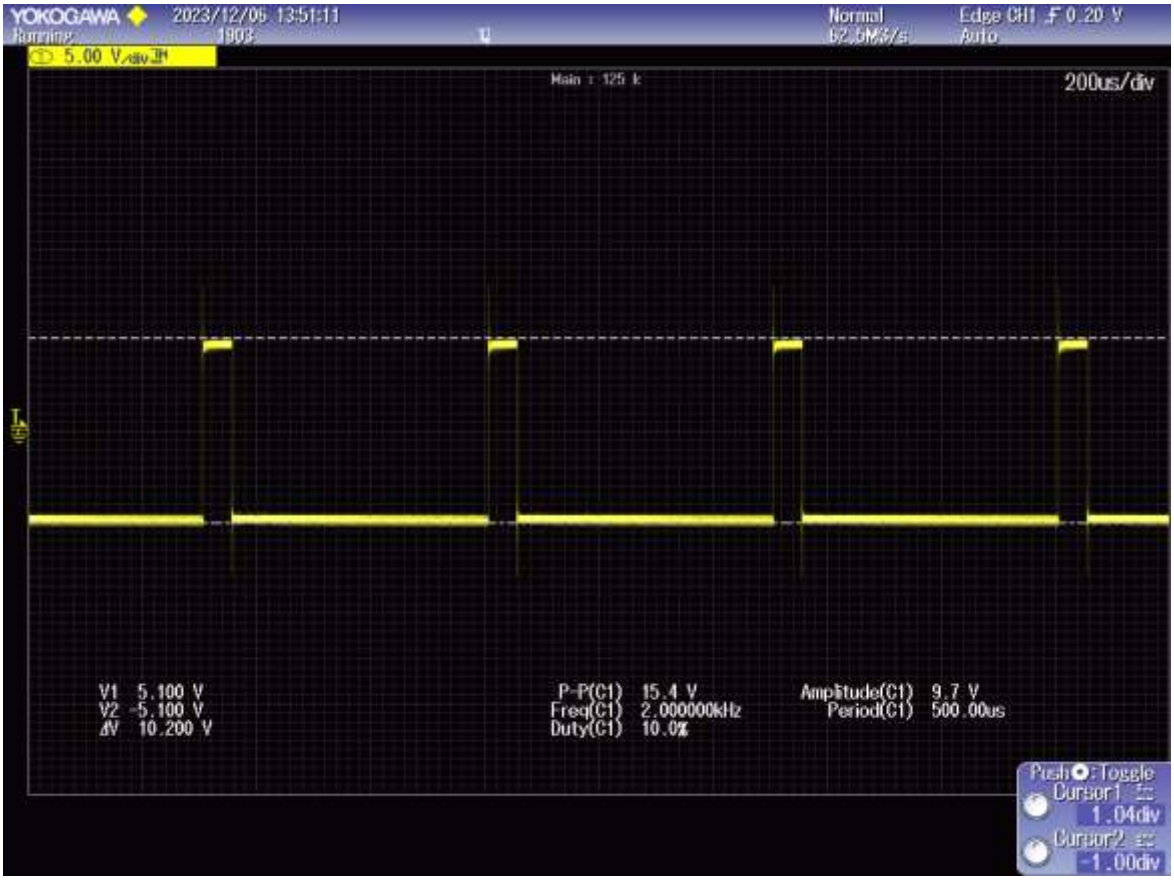


Figure 5: Observed Time domain square waveform-10% Duty cycle

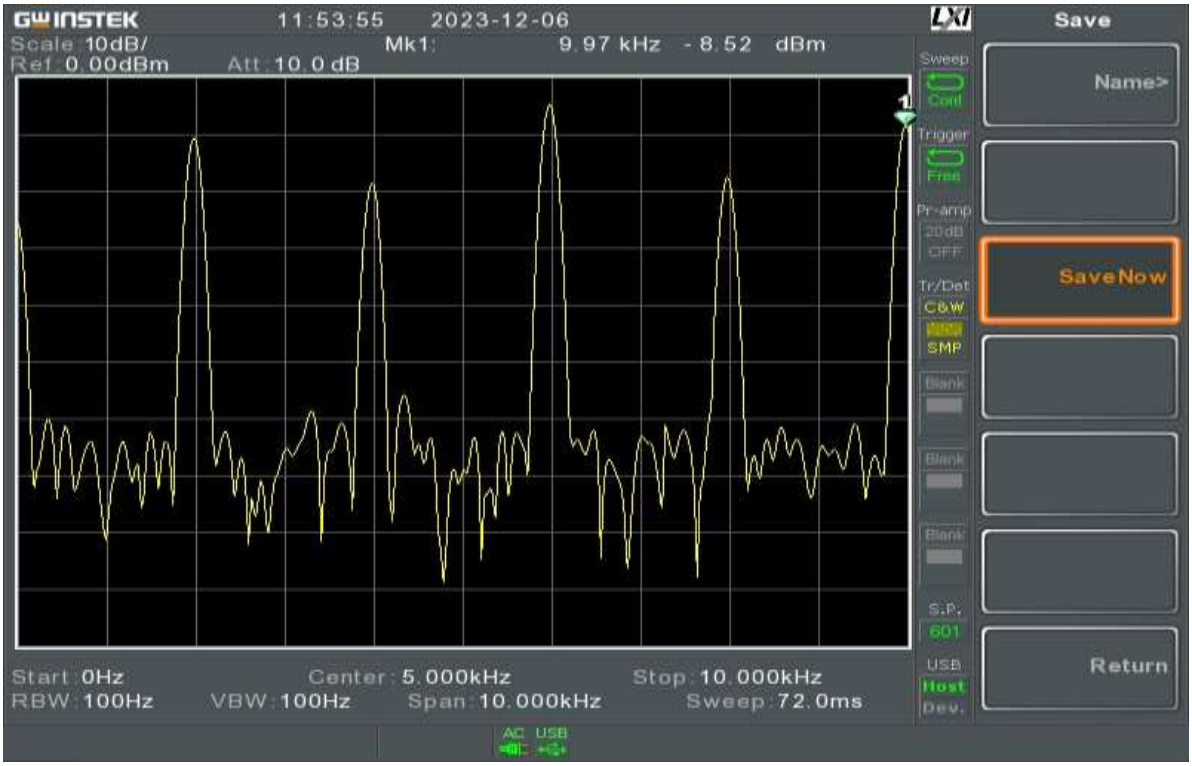


Figure 6 : Observed Frequency domain waveform-10% Duty cycle

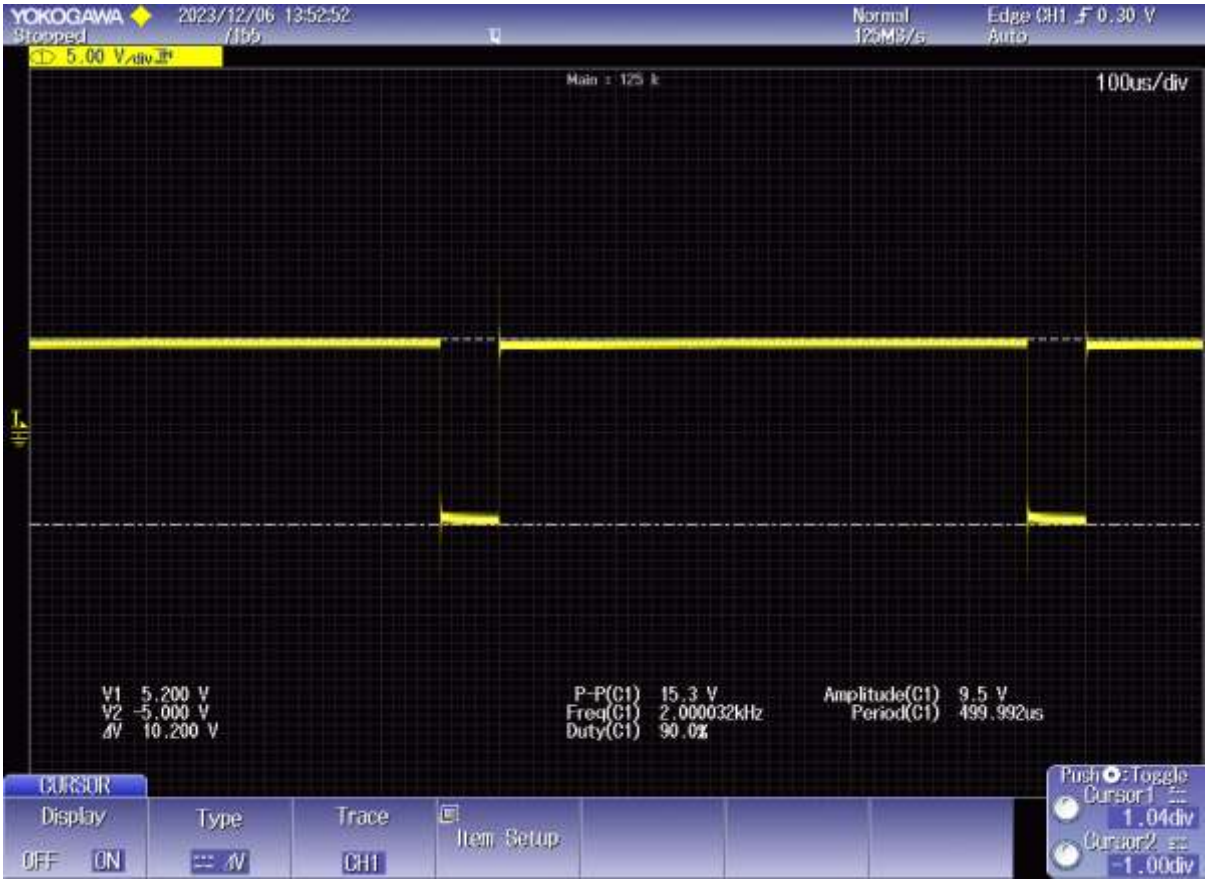


Figure 7 : Observed Time domain square waveform-90% Duty cycle

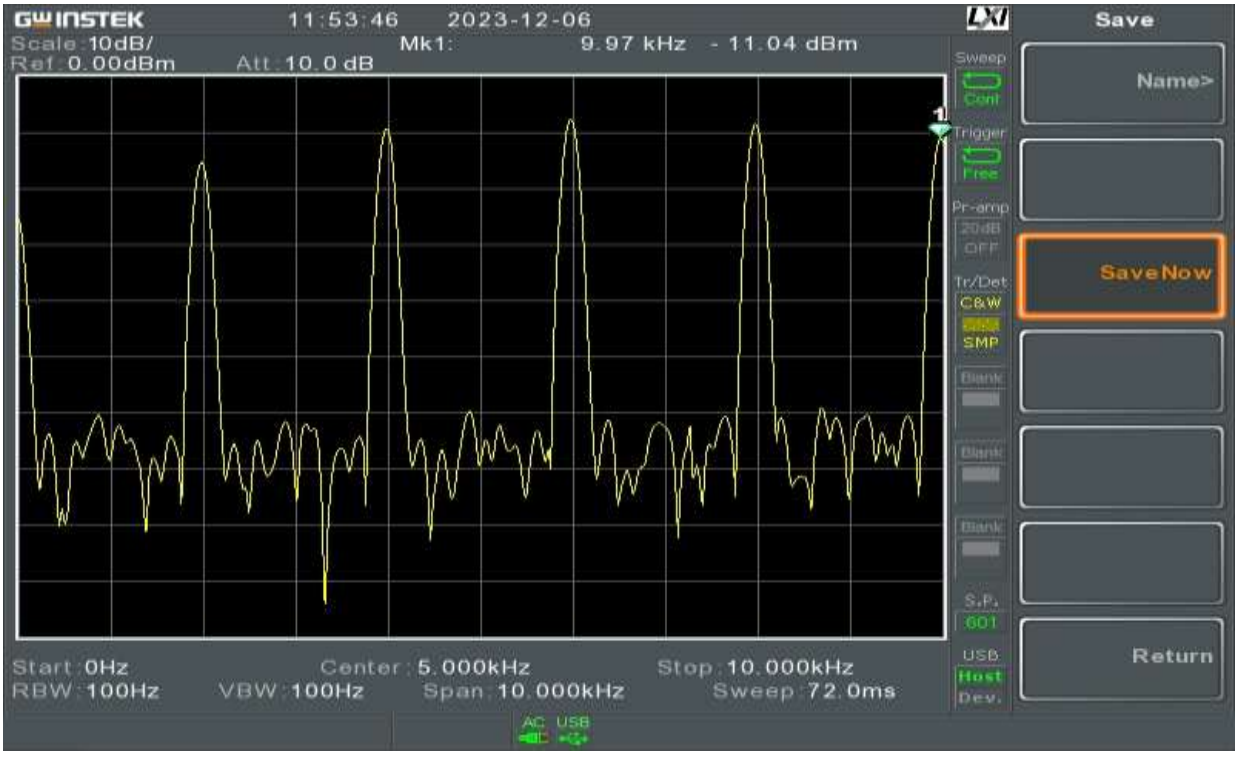


Figure 8 : Observed Frequency domain waveform-90% Duty cycle

1.2 Part 02: Observe the Lissajous pattern using oscilloscope while comparing frequency and the phase shift of the different waveform.

1

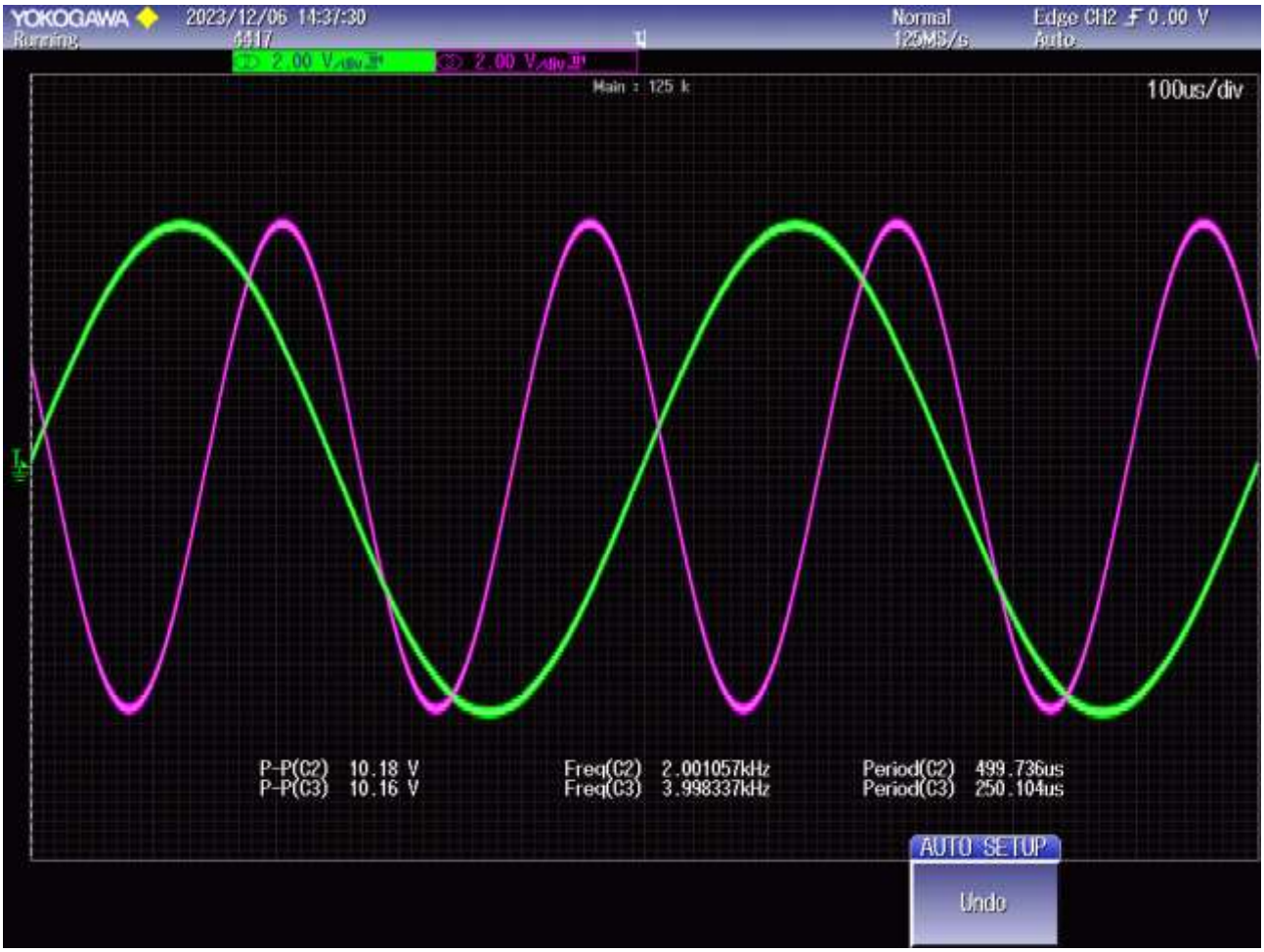


Figure 9 : Generated Teo sinusoidal waveform in Time domain

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- Phase change-0^0

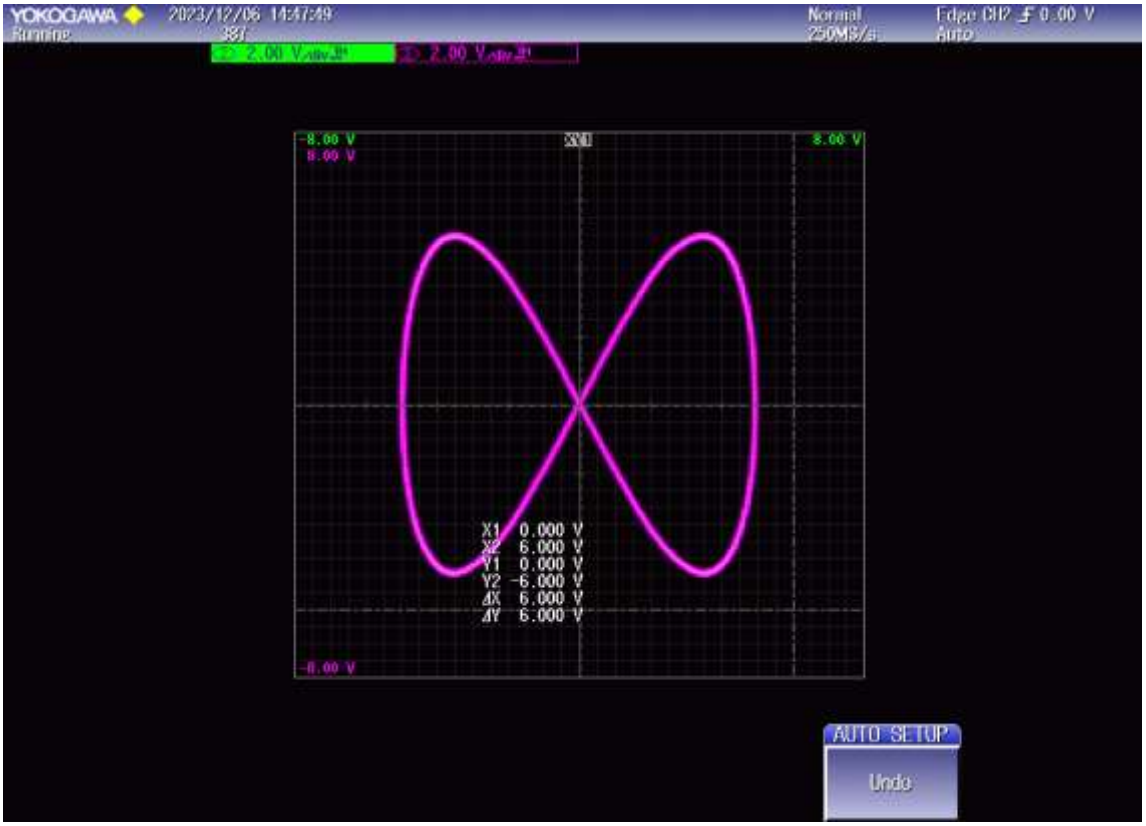


Figure 10 : Observed Lissajous pattern for phase change-0^0

- Phase change-45^0

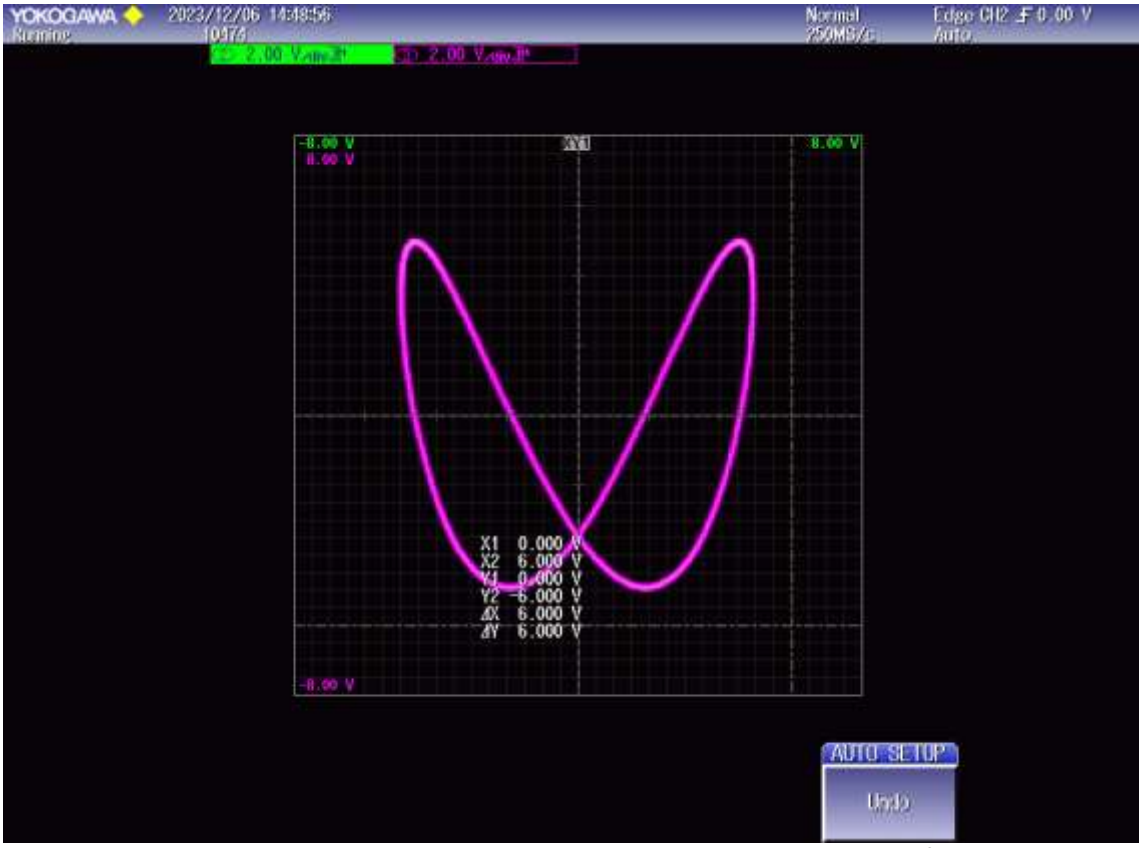


Figure 11 :

Observed Lissajous pattern for phase change-45^0

- Phase Change-90^0

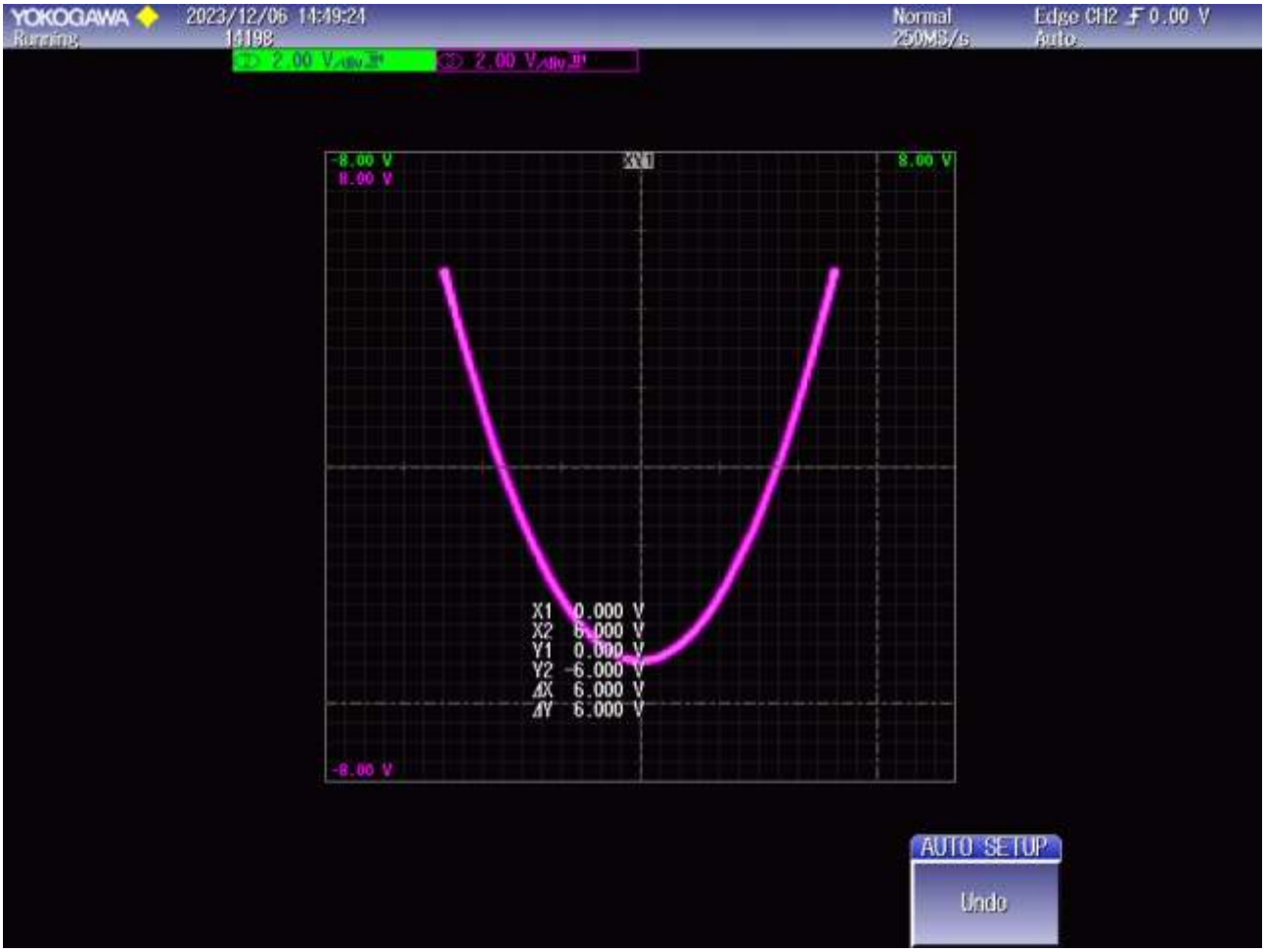


Figure 12 : Observed Lissajous pattern for phase change-90^0

2 Calculations

2.1 Specimen calculations

$$\text{Frequency Ratio}=\frac{f_x}{f_y}=\frac{Number\ of\ horizontal\ crossing}{Number\ of\ vertical\ crossing}$$

$$\text{Frequency Ratio}=\frac{4kHz}{2kHz}=2$$

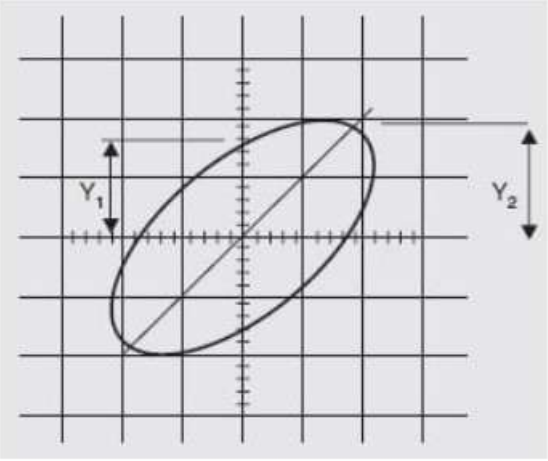
Considering Figure 11,

$$\text{Experiment frequency ratio}=\frac{4}{2}=2$$

3 Discussion

1) Explain the phase angle and magnitude relationships of two sinusoids of the Lissajous pattern by using examples.

We can apply two waveforms to horizontal and vertical amplifiers. We can apply waveforms to both horizontal and vertical amplifiers by using Oscilloscope. Then Oscilloscope will display a pattern that pattern call Lissajous pattern. This pattern formed when two sinusoids applied to vertical and horizontal amplifiers. The frequency and phases are different in those two sinusoids.



$\sin\theta = \frac{Y_1}{Y_2}$

$\theta = \sin^{-1}(\frac{Y_1}{Y_1})$

Y_1=8units

Y_2=10units

$\theta = \sin^{-1}(0.8)$

$\underline{\theta = 53.13^0}$

$\frac{f_x}{f_y} = \frac{Numbr\ of\ horizontal\ crossing}{Number\ of\ vertical\ crossing}$

$= \frac{2}{2}$

$= \underline{1}$